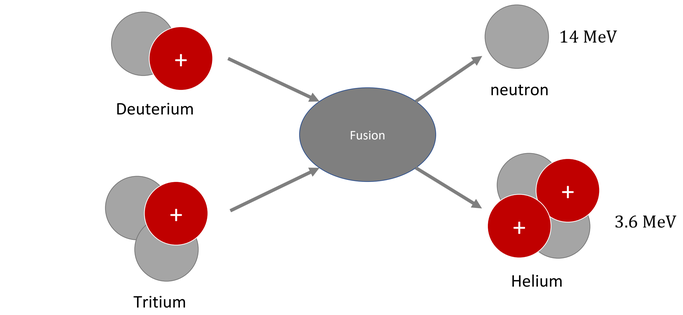
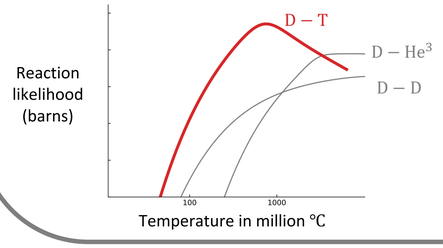
ARC

Affordable Robust Compact tokamak

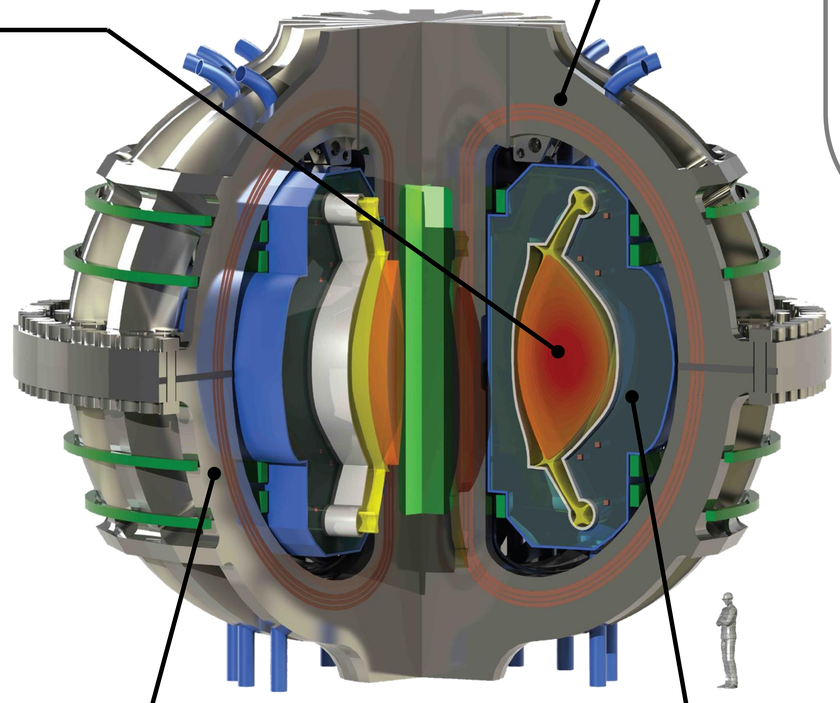
Sorbom et al (2015), Fus. Eng. Design,
DOI: 10.1016/j.fusengdes.2015.07.008

What is Deuterium-Tritium fusion?





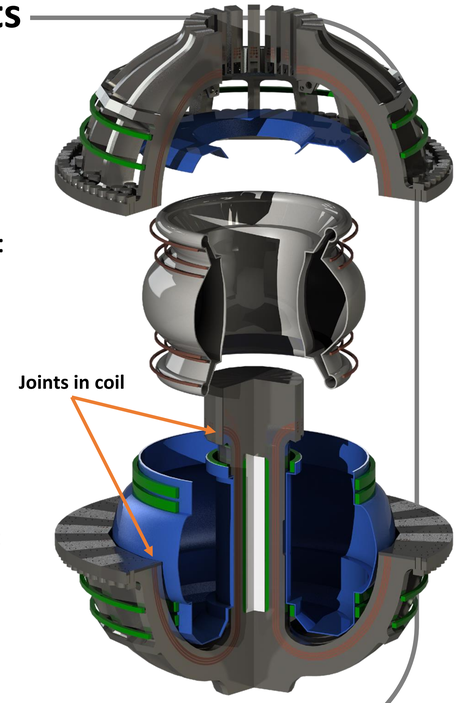
Why do we need Deuterium and Tritium?



Demountable magnets

Using **H**igh **T**emperature **S**uperconductors enables joints:

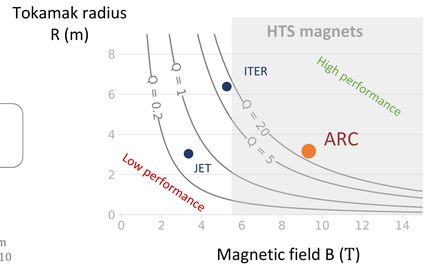
- ☑ Magnets can be mounted/demounted many times
- ☑ Easier maintenance from the top
- ☑ Vacuum vessel can be replaced and is no longer a lifetime component

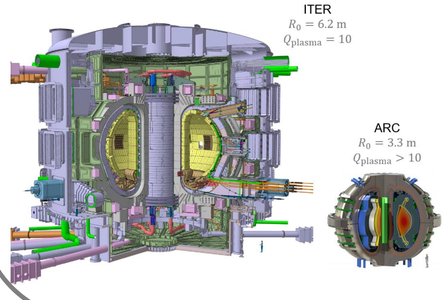


High Temperature Superconducting Magnet

Increasing the magnetic field greatly improves fusion performance

$$Q_{\text{plasma}} = \frac{\text{fusion power}}{\text{heating power}} \propto R^3 \boldsymbol{B}^5$$



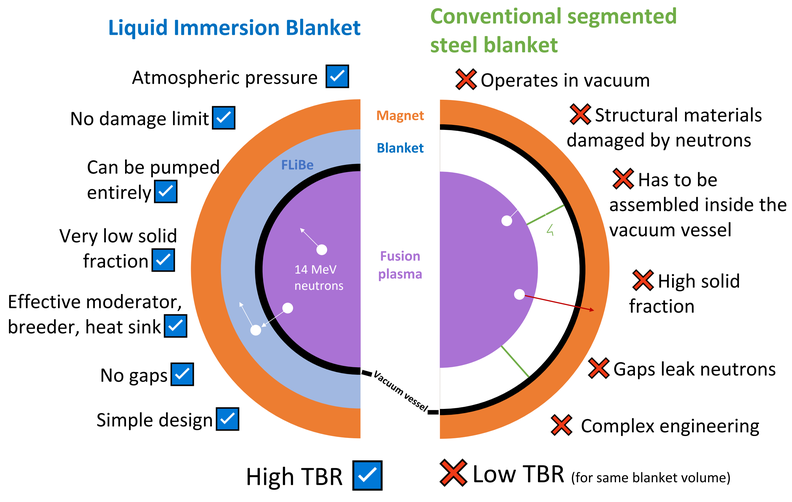


Using HTS magnets allows tokamaks to be more compact and cheaper

Major radius	3.3 m
TBR	1.1
Fusion power	525 MW
Multiplication factor *Q*	>10
Magnetic Field	9.2T

Liquid Immersion Blanket

Breeding tritium | Shield from neutrons | Extract heat



Why Lithium?

$$D + T \rightarrow He + n$$
$$Li + n \rightarrow T + He$$

Why Beryllium?

$$^9Be + n \rightarrow {}^8Be + n + n$$

$$\text{Tritium Breeding Ratio} = \frac{\text{Tritium produced}}{\text{Tritium consummed}}$$

Theoretical sufficiency → TBR = 1
Losses + start other plants → **TBR ≈ 1.1 − 1.2**

LIBRA

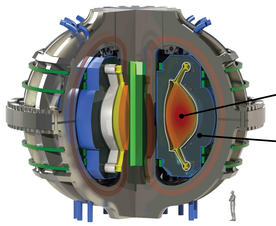
Liquid **I**mmersion **B**lanket - **R**obust **A**ccountancy

MIT Plasma Science and Fusion Center

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Tritium breeding

2030s



The ARC concept relies on the Liquid Immersion Blanket (LIB) for breeding tritium

$$D + T \rightarrow He + n$$

$$Li + n \rightarrow T + He$$

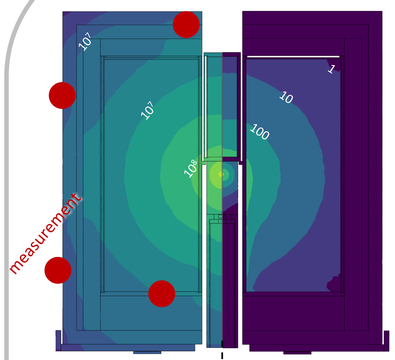
$$\text{Tritium Breeding Ratio} = \frac{\text{Tritium produced}}{\text{Tritium consummed}}$$

Can we test this concept now?

Goals of LIBRA

- ✅ To demonstrate the concept of LIB (TBR ≈ 1)
- To improve the Technology Readiness Level of LIB
- To validate numerical models for tritium transport
- To gain knowledge on molten salt behaviour in fusion environments

Neutronics



Neutron flux $n/cm^2/s$ | **Tritium production** $nCi/l/h$

- Neutronics simulations track the path of neutron particles
- Used to predict tritium generation, neutron flux, heating, production of activated materials…

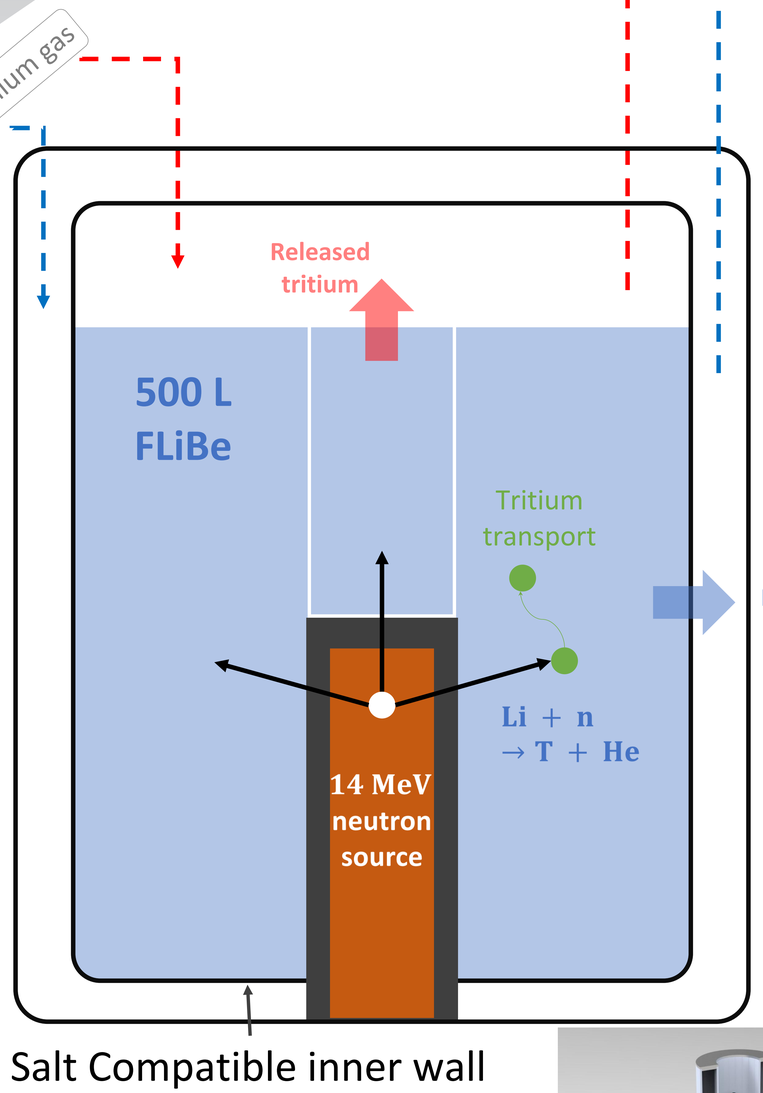


Activation foils

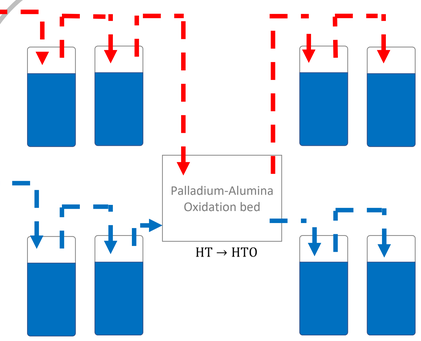
- LIBRA will provide neutronics data for validating codes
- Activation foils, liquid organic scintillators and diamond detectors as used to characterize the neutron source and the experiment



Diamond detection

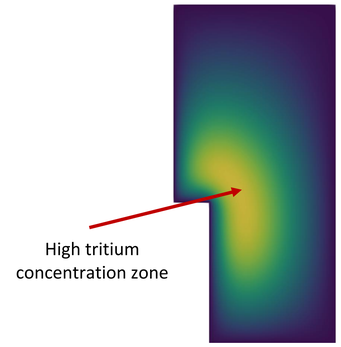


Tritium detection



Water bubblers

- Bubblers with oxidation furnace capture different chemical forms of tritium
- Salt radiochemistry impacts tritium forms and transport dynamics
- LIBRA will provide data for simulation codes validation
- The impact of sparging bubbles in the FLiBe will also be studied



Tritium transport can be modelled at full scale

LIBRA Pi

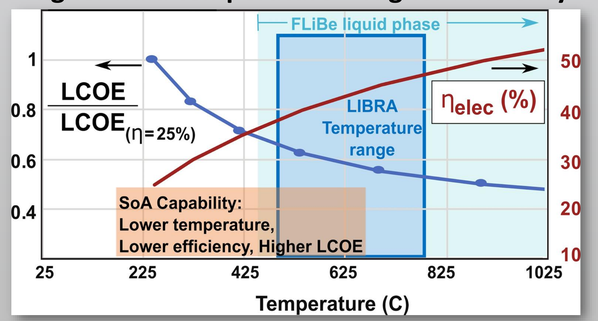
LIBRA ONE specs

Salt volume	700 L
TBR	1.05
Temperature	700C
DT Neutron energy	14.1 MeV



LIBRA ONE

Higher fluid temperature = Higher Efficiency



BABY

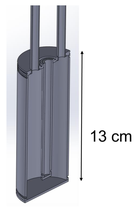
Build A Better Yield blanket

MIT Plasma Science and Fusion Center

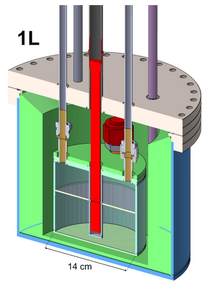
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Scaling up

100 mL



1L

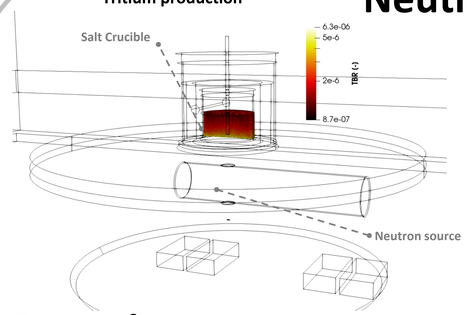


BABY specs

Salt volume	0.1-1 L
TBR	~10^{-3}
Temperature	700C
DT Neutron energy	14.1 MeV

Neutrons

OpenMC Tritium production



Neutronics

- Neutronics simulations track the path of neutrons
- Used to predict tritium generation, neutron flux, heating, production of activated materials…

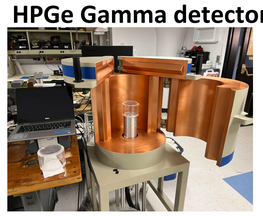
Detection

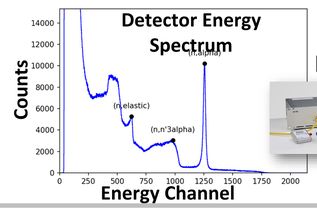
- LIBRA will provide neutronics data for validating codes
- Activation foils and diamond detectors as used to characterize the neutron sources and track the neutron flux

Activation foils



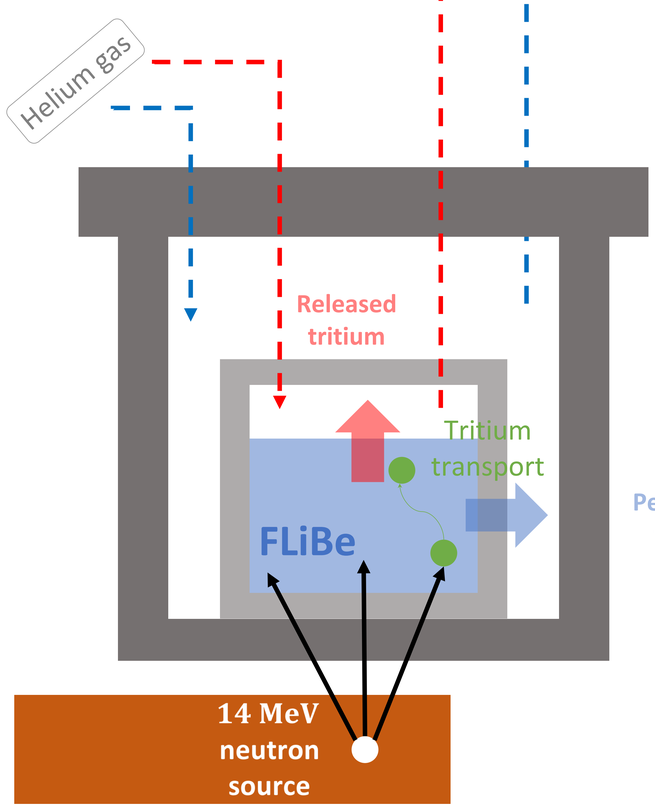
HPGe Gamma detector





Diamond detector



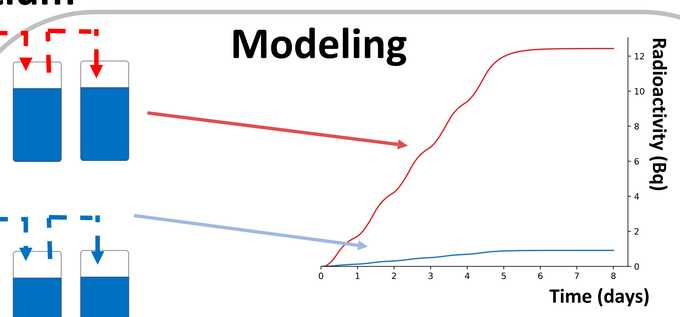


1L setup



Tritium

Modeling



0D tritium transport model:

$$V\frac{\partial c}{\partial t} = S - A_{\mathrm{p}}\ k_{\mathrm{p}}\ c - A_{\mathrm{r}}\ k_{\mathrm{r}}\ c$$

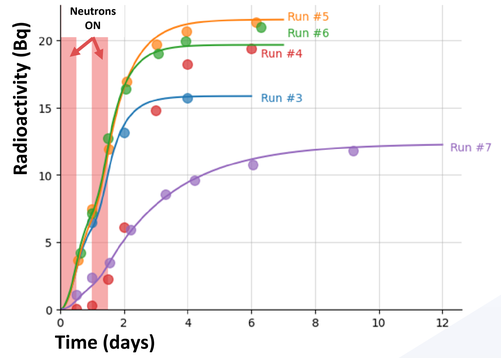
V: salt volume
c: salt tritium concentration
S: tritium source
A_i: surface area
k_i: mass transport coefficient

Detection

Tritium water bubblers

- Bubblers collect 99% of the tritium
- Water samples are analyzed with Liquid Scintillation Counting
- Measurements on BABY serve as validation data

0.1 mL BABY Top release measurements



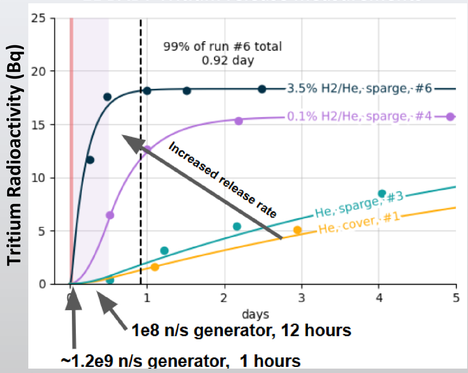
100ml results

- Tritium breeding
- Validated neutronics and tritium transport models
- Reproducibility
- Salt chemistry impacts on tritium dynamics

1L BABY specs

- 10x Higher tritium breeding ratio
- Wall-permeated tritium
- Accelerated tritium dynamics
- Quantify and refine uncertainties

1L BABY Tritium release measurements



1L BABY results

- Scaled up tritium breeding
- Validated neutronics and tritium transport models
- Informed tritium dynamics model parameters (volume, breeding rate)